Learning Objectives

- To learn about the various grades of lubricating oils used (SAE 20, SAE 40, SAE 90, SAE 120) in cars, buses and trucks.
- To learn the importance of engine oil and lubrication system to avoid friction and overheating.

8.0 INTRODUCTION

The process of using lubrication oil to reduce friction between two moving parts is called as lubrication. Metallic contacts between two moving pats lead to friction, heat, wear, noise and seizure of the engine. To overcome this problem, lubricating system is needed in all automobiles. Lubrication is also helpful for smooth functioning of the moving parts in contact. Automobiles have many moving and rotating parts. If lubrication system is not present in vehicles, the durability of the components will be reduced due to wear and tear caused by friction. Hence to increase the life time of the engine, lubrication system is essential in automobiles.

8.1 ADVANTAGES OF LUBRICATION

The Engine Lubrication system

1. Reduces the friction between moving parts.
2. Reduces the damage of parts due to moving in contact.
3. Removes/Decreases the heat due to friction.
4. Cleans the tiny metal particles and dust that are present between two moving parts.
5. Acts as a seal between piston ring and cylinder to provide compression.
6. Reduces the vibrations and makes the parts of the engine to function silently.
7. Increases the strength of engine.
8. Prevents rusting.
9. Helps in increasing the lifetime of engine components.

8.2 PROPERTIES OF LUBRICATING OIL

The lubricating oil or paste used for reducing friction is known as lubricant. The related properties of the Engine Lubricants are



1. Viscosity.
2. Oiliness or Adhesiveness.
3. Fluidity.
4. Volatility.
5. Flash point.
6. Fire Point.
7. Stability.
8. Corrosiveness.
9. Cleanliness.
10. Emulsification.
11. Cloud point.
12. Foaming.

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